

README — novaexopia/

The "Claw" Harness Runtimes, 3-APK Topology & Terminal Shells

W5+H Subsystem Identity

- **WHO:** Houses **Horizons UI** (APK 1), **Æsc** (APK 2: Terminal daemon), **Æyre** (APK 3: Media daemon), and OpenWiki TUI.
- **WHAT:** Native Android services, Ktor WebSocket bridges, ADB loopback clients, Prime Agent RLM loops, and MCP connectors.
- **WHEN:** Active continuously from Android boot (**START_STICKY**); listens for UI interactions, shell commands, and audio streams.
- **WHERE:** [novaexopia/](#) (Federated Subsystem Root).
- **WHY:** Eliminates Android sandbox restrictions and Low Memory Killer (LMK) process termination, bypassing Termux entirely.
- **HOW:** Decouples UI rendering from native execution daemons, streaming data across local UNIX sockets (**AF_UNIX**) and Netty WebSockets.

Internal Directory Topology

novaexopia/

```
|— README.md                # This document (APK architecture & harnesses)
|
|— manifest.jsonl           # Local cryptographic catalog of runtimes & tools
|
|— horizons-ui/             # APK 1: Master Visual Presentation Shell
|   |— app/src/main/assets/web/    # Chromium WebView HTML5 / React bundle
|   |— ui_tiles/                # Chat interface tile, Terminal visualizer, Model picker
|   |— bridge/                  # Persistent WebSocket clients to Æsc and Æyre
|
|— aesc/                     # APK 2: System Terminal Daemon (Termux Replacement)
```

- | | — src/main/java/com/horizons/ui/adb/ # WirelessAdbBridgeService.kt (Loopback on 127.0.0.1:5555)
- | | — npu_watchdog/ # npu_manager.py (UNIX socket hot-swap daemon)
- | | — scripts/ # system_housekeeper.sh, agent_panel.sh
- |
- | — aeyre/ # APK 3: Media & Sensory Ingress Daemon
- | | — voice_stack/ # Silero VAD, Moonshine ONNX STT, Kokoro-82m TTS
- | | — screen_vision/ # Android Media & Video Game SDK frame capture
- |
- | — openwiki-tui-harness/ # Interactive Terminal Workspace Shell
- | | — config_profiles/
- | | | — file_administrator.yaml # Profile A: Master database housekeeper
- | | | — oracle_helpdesk.yaml # Profile B: On-device interactive help desk
- | | — src/ # OpenWiki CLI terminal bindings
- |
- | — modular_harnesses/ # Agent Execution Frameworks
- | | — prime-agent/ # Prime Intellect RLM loop inside persistent Python REPL
- | | — ecc-claude-code/ # Everything Claude Code specialized prompt skills
- |
- | — mcp_connectors/ # Model Context Protocol Server Suite
- | | — filesystem_mcp/ # Node.js @modelcontextprotocol/server-filesystem
- | | — sqlite_mcp/ # Python on-device embedded key-value server

└─ postgres_mcp/

Remote OB1 connection over Tailscale

Beginner-Proof Implementation Rules

1. **Fault Isolation:** If the model process dies inside `aesc/`, `horizons-ui/` remains completely stable and reconnects automatically via WebSockets.
2. **No Root Required:** All shell privileges are obtained through the onboard loopback connection to `127.0.0.1:5555` via Wireless Debugging.